

Ultrafast Electro-Absorption Modulators and Fin-Photodiodes in a SiGe Photonics Platform

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Abstract We present C-band SiGe electro-absorption modulators with an RC-time-constant-estimated 3-dB bandwidth of 192 GHz and a nearly flat measured response up to 110 GHz. Monolithic co-integration with 200-GHz SiGe fin photodiodes provides an unprecedented combination of high-performance devices, paving the way towards 400-Gbaud optoelectronic systems. ©2026 The Authors

Introduction

The explosive growth in AI model complexity has pushed the demand for mass-manufacturable, high-speed optical interconnects to unprecedented levels. As model sizes and cluster scales grow, inter-chip bandwidth increasingly bottlenecks system performance, making low-cost, high-volume photonic solutions a critical enabling technology. Silicon photonics has emerged as the leading platform to meet this demand, leveraging mature CMOS fabrication infrastructure to deliver transceivers at scale. Industrial silicon photonic platforms still predominantly rely on silicon modulators based on the plasma-dispersion effect [1,2], for which operation beyond 100 GHz remains challenging due to the inherent bandwidth-modulation efficiency trade-off. Consequently, modulator solutions enabling bandwidths well beyond 100 GHz are highly sought after. To address this need, alternative modulator concepts based on III-V materials, LiNbO₃, LiTaO₃, silicon-organic hybrid integration, and plasmonics have been explored [3–7]. However, these approaches typically introduce substantial process complexity and may raise material-contamination concerns in CMOS-compatible fabrication environments. In recent years Ge-rich SiGe electro-absorption modulators (EAMs) utilising the Franz-Keldysh effect (FKE) have

emerged as a promising device class, achieving >110 GHz 3-dB bandwidth and offering full CMOS-compatibility for a seamless monolithic integration on silicon [8,9].

In [9] we previously reported the monolithic co-integration of Ge-rich SiGe EAMs with 3-dB bandwidth of approximately 140 GHz, together with fin-photodiodes (PDs), setting a benchmark for the highest performing combination of native silicon photonic modulators and photodiodes within one platform. However, these first-generation devices exhibited relatively high insertion loss and limited extinction ratio across the C-band.

Here, we present an advanced SiGe photonics platform featuring next-generation SiGe EAMs with high extinction ratio across the entire C-band, an RC-time-constant-estimated 3-dB bandwidth of 192 GHz, and a nearly flat measured S_{21} response up to 110 GHz. The performance improvement is enabled by:

- advanced atomic-layer epitaxy for improved Si homogeneity in the SiGe region,
- optimized Si content for low-loss operation and high extinction ratio across the C-band,
- refined doping profiles in the in-situ-doped polysilicon layers.

Importantly, the co-integrated >200-GHz SiGe fin photodiodes retain their performance. To the best of our knowledge, this combination of ultrahigh-speed modulation and photodetection is unprecedented in a monolithic photonics platform.

Device concept

The electro-absorption modulator as well as the monolithically co-integrated fin-photodiode consist of a rectangular SiGe-region sandwiched between two complementary in-situ doped poly-silicon regions. The in-situ doped silicon layers adjacent to the SiGe-region form lateral extensions that enable in combination with the formation of a nickel-platinum-silicide layer low-resistance contacts for accessing the device. To enable C-band operation we have introduced around 1.5% Si

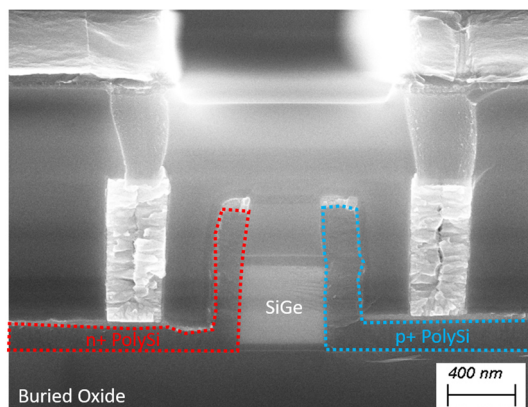


Fig. 1 Cross-sectional SEM-image of SiGe electro-absorption modulator.

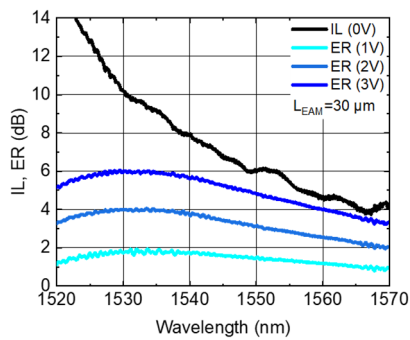


Fig. 2 Static extinction ratio at various reverse biases and insertion loss of SiGe EAMs. The measurements were performed at an optical power of -2.5 dBm at the optical fibre tip.

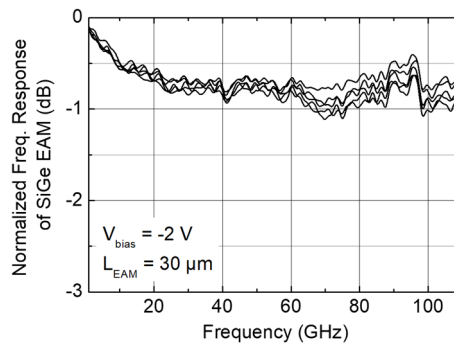


Fig. 3 Normalised frequency response of SiGe EAM at $\lambda=1550$ nm, from five chips measured on one wafer.

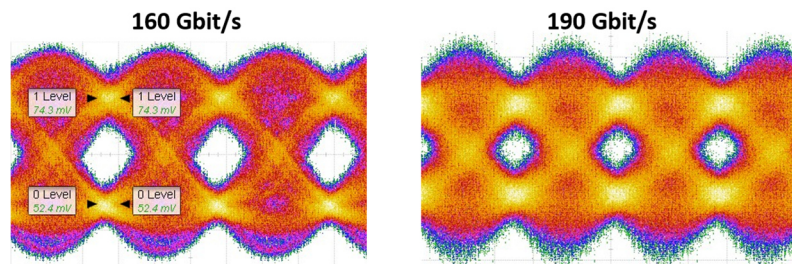


Fig. 4 Eye diagrams of RF-probed 30 μm long SiGe EAM at data rates of 160 and 190 Gbit/s ($\lambda=1550$ nm). The dynamic extinction ratios are 1.52 dB (160 Gbit/s, -2 V, 4-tap Rx equalization) and 0.8 dB (190 Gbit/s, -1.3 V, 8-tap Rx equalization), respectively. The AWG output signal was 2.4 V_{pp}. Due to RF-losses at the probes we estimate 1.8 V_{pp} at most at the EAM.

into the Ge by using a novel atomic layer epitaxy process, building upon the process discussed in [10]. The geometry of the devices can be optimized independently by layout changes [9,11]. The EAMs benefit from a rather wide SiGe region with a width in the range of 450 nm for sufficient mode confinement and moderate propagation losses, whereas the photodiodes are patterned down to a width in the range from 100 nm to 300 nm to enable shortest drift times for the photogenerated charge carriers. The EAM has a length of 30 μm , while the PDs are 10 μm long. A SEM-image is shown for the cross-section of a 450 nm wide SiGe EAM is shown in Fig. 1.

Results and Discussion

Insertion loss and extinction ratio of the 30 μm long EAM are illustrated in Fig. 2. The static measurements were performed at a low optical power of -2.5 dBm at the fibre tip to avoid self-heating effects, resulting in an increased static ER that would not translate to a higher dynamic ER [12]. At $\lambda=1550$ nm the EAM yields an insertion loss of 6 dB and an ER of 5 dB at -3 V. The substantially increased ER compared to our prior device generation from [9] can be attributed to our refined SiGe epitaxy process, achieving a more homogenous silicon distribution. Further the Si-concentration was adjusted to 1.5% to now enable high extinction ratio operation in the whole C-band while maintaining low to moderate optical

losses. The RF-behaviour of the EAMs was characterized using a 110-GHz Lightwave Component Analyzer (LCA). The EAMs show a nearly flat electro-optical frequency response up to 110 GHz (Fig. 3). Due to the measurement setup limitations an exact determination of the 3-dB bandwidth is not possible. However, as the bandwidth of FKE-based EAMs is governed by the RC-time constant we estimate the 3-dB bandwidth of the device based on electrical small-signal measurement extracted series resistance and capacitance. At -3 V bias the EAM exhibits a series resistance of 57.5 Ω and capacitance of 7.7 fF. Considering the 50 Ω impedance of the measurement system this corresponds to an RC-limited 3-dB bandwidth of 192 GHz.

Eye-diagrams of the EAM were measured at NRZ data rates of 160 Gbit/s and 190 Gbit/s (Fig. 4). The EAMs were characterized using a probe station with a 110-GHz RF probe. Electrical NRZ signals were generated by an arbitrary waveform generator, with a 110-GHz bias tee supplying reverse bias from a source/measure unit. Light from an external cavity laser was modulated by the EAM and amplified by an erbium-doped fibre amplifier and filtered to remove excess noise. The signal was then detected by a 100-GHz photodetector connected to a sampling oscilloscope. Open eye diagrams were achieved using only minor linear equalization and smoothing on the oscilloscope.

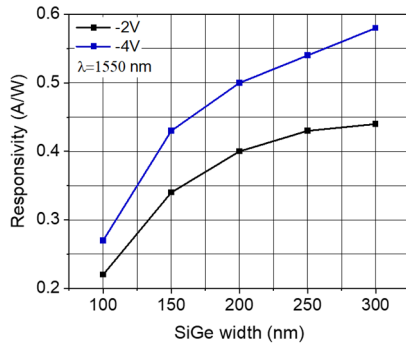


Fig. 5 Internal responsivity of SiGe PD (referred to optical power in the Si waveguide after the grating couplers). The measurements were performed for various fin-widths at -2 V and -4 V bias at $\lambda=1550$ nm.

The demonstrated SiGe fin-photodiodes were manufactured on the same wafer as the SiGe EAMs, which enables entire transceiver applications within a single technology platform. No separate epitaxy steps were required for the fabrication of the EAMs and the PDs. We present PDs with SiGe widths from 100 nm to 300 nm and a length of 10 μm . In Fig. 5 we provide internal responsivity values, relating the generated photocurrent to the optical power within the silicon waveguide leading to the photodiode, after subtracting the losses of the grating couplers. The SiGe-fin PDs show similar responsivity values as the pure Ge PDs [11, 13], which can be attributed to the Franz-Keldysh effect, increasing the absorption within the SiGe with an applied electric field. We also conducted RF-measurements on the 110 GHz LCA for the photodiodes. As illustrated in Fig. 6, all PDs have a 3-dB bandwidth beyond 110 GHz. The two narrowest variants have a S_{21} roll-off of less than 0.5 dB at 110 GHz, similar to the fin-PDs reported in [13], that achieved 240 and 265 GHz bandwidths, confirmed by heterodyne measurements.

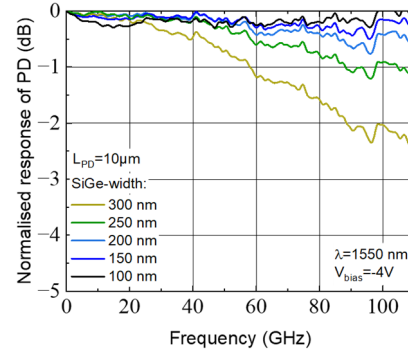


Fig. 6 Normalised frequency response of SiGe-PDs with fin widths ranging from 100 nm to 300 nm, measured at $\lambda=1550$ nm with -4 V bias.

To benchmark our devices, we compare them against recently reported state-of-the-art Franz-Keldysh-effect-based C-band Ge and SiGe electro-absorption modulators, as well as co-integrated photodiodes where applicable (Table 1). In terms of bandwidth, our EAMs and PDs outperform previously reported silicon photonic devices.

Conclusion

We demonstrated the monolithic co-integration of ultrafast SiGe electro-absorption modulators and SiGe fin photodiodes. Fabrication was carried out entirely in IHP's 200-mm BiCMOS pilot line. The EAMs exhibit an RC-time-constant-estimated 3-dB bandwidth of 192 GHz, while the measured electro-optical response remains nearly flat up to the 110-GHz limit of the measurement setup. Together with fin photodiodes exceeding 200 GHz bandwidth, this monolithic integration combines exceptionally high-speed modulation and photo-detection on a single SiGe photonics platform, paving the way towards systems operating at symbol rates of 400 GBaud.

Table. 1 Benchmark of state-of-the-art Ge and SiGe Franz-Keldysh-Effect electro-absorption modulators and co-integrated photodiodes at C-band wavelength.

	Ref	imec [14] (2016)	imec [8] (2025)	PETRA [15] (2023)	IHP [9] (2025)	IHP (this work)
EAM	Length (μm)	40	49.6	40	40	30
	Mod. 3-dB BW (GHz)	>50 (85 ^a)	>110 (140 ^a)	>67 (85 ^a)	100	>110 (192 ^b)
	Insertion Loss (dB)	4.4 (1560 nm)	3.6 (1540 nm)	5 (1550 nm)	10 (1580 nm)	6 (1550 nm)
	Static ER	7.8 (1550 nm, -3V)	3.4 (1540 nm, -2V)	3 (1550 nm, -3V)	4.5 (1580, -3V)	5 (1550 nm, -3V)
	Dynamic ER (dB)	3 (2V, 50G)	n/a	1.5 (0.8V, 100G)	1.6 (1.8 V, 160G)	1.52 (1.8 V, 160G) 0.8 (1.8 V, 190G)
	Symbol rate (GBaud)	50 (NRZ)	224 (PAM-4)	100 (NRZ)	160 (NRZ)	190 (NRZ)
PD	PD 3-dB BW (GHz)	>50	-	60	>200 ^c	>200 ^c
	Int. Responsivity (A/W)	0.86 ^d (-1V, 1550nm)	-	0.83 (-4V, 1550 nm)	0.27 (-4V, 1580 nm)	0.42 (-4V, 1550 nm)

^a Estimation based on extrapolated S_{21} -curve, ^b RC-time-constant-estimated, ^c S_{21} frequency response roll-off < 0.5 dB at 110 GHz similar to the heterodyne measurement results of the 240 and 265 GHz fin-PDs reported in [13],

^d Responsivity provided for Ge PD, no co-integrated SiGe-based PD reported

Acknowledgements

We gratefully acknowledge support from IHP's entire cleanroom staff and David Stolarek for taking SEM images. We also acknowledge funding by the projects DFG-EPIC-COM (ZI1283-12-1), HE SPACELINK (101189988), HE QOSILCIOUS (101187539) and QuantERA QISSME (101017733)

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